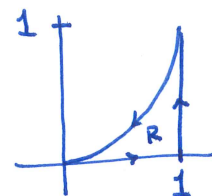


Green's Theorem

SOLNS

Problem 1

$$a) \vec{F} = \overbrace{2xy^3}^M \vec{i} + \overbrace{4x^2y^2}^N \vec{j}$$



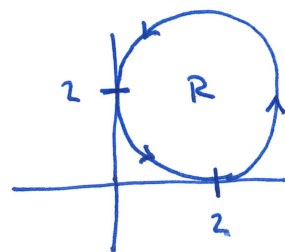
$$N_x - M_y = 8xy^2 - 6xy^2 = 2xy^2$$

$$\begin{aligned} \oint_C \vec{F} \cdot d\vec{r} &= \iint_R 2xy^2 dA = \int_0^1 \int_0^{x^3} 2xy^2 dy dx \\ &= \int_0^1 \left. \frac{2}{3}xy^3 \right|_0^{x^3} dx \\ &= \int_0^1 \frac{2}{3}x^{10} dx \\ &= \left. \frac{2}{33}x^{11} \right|_0^1 = \boxed{\frac{2}{33}} \end{aligned}$$

$$b) \vec{F} = \overbrace{[4x-2y]}^M, \overbrace{[2x-4]}^N$$

$$N_x - M_y = 2 - (-2) = 4$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R 4 dA = 4 \cdot \text{Area}(R) = 4(4\pi) = \boxed{16\pi}$$



Problem 2

$$\frac{1}{2} \oint_C -y dx + x dy$$

Note: $\vec{F} = [-y, x]$

$$N_x - M_y = 1 - (-1) = 2$$

$$= \frac{1}{2} \iint_R 2 dA$$

$$= \frac{1}{2} (2 \cdot \text{Area}(R))$$

$$= \text{Area}(R) \quad \checkmark$$